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From points to waves: Fast ocean wave spatial–temporal fields estimation using ensemble transform Kalman filter with optical measurement

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ABSTRACT

Accurate spatial–temporal wave measurement is vital for ocean engineering applications. Although stereo vision shows great potential in this field, performing dense reconstruction requires processing vast amounts of pixel data, which reduces the efficiency of stereo image matching and subsequent point cloud processing. Recently, the paradigm of fusing sparse 3D points with predictions has emerged as a promising solution that balances accuracy and efficiency, yet requires an optimization framework capable of handling both state estimation and robust outlier filtering. Therefore, this study proposes a Kalman filter (KF)-based method for ocean wave field estimation, aiming to improve efficiency through recursion and to remove outliers and interpolation errors in measurements. The method leverages linear gravity wave dispersion relations for prediction, with sparse 3D points interpolated to a uniform grid as measurements. To address limitations of high-dimensional data processing, the study implements the Ensemble Transform Kalman Filter (ETKF), incorporating fuzzy logic to handle potential outliers. By maintaining an ensemble of states and employing ensemble transformation techniques to avoid computationally expensive matrix inversions, ETKF significantly improves recursive processing efficiency. Both CPU and GPU implementations were evaluated on published and field-collected datasets, demonstrating superior performance in efficiency, accuracy, and robustness compared to existing methods under the same paradigm.

1. Introduction

Accurate spatial–temporal wave measurement (Benetazzo et al., 2016; Li et al., 2022) is essential for ocean engineering applications, ranging from navigation to offshore structure safety, and wave energy harvesting. Traditional single-point measurement systems cannot adequately capture critical features of three-dimensional sea surfaces, including directionality, wave height, and surface slope. Furthermore, research on spatial–temporal fields based on three-dimensional wave measurements has yielded new insights into wave modeling (Leckler et al., 2015; Barbariol et al., 2017). Consequently, developing efficient and robust spatial–temporal wave measurement methods has become a research priority.

Wave measurement techniques have evolved significantly over the past decades. Traditional methods relied on wave gauges and surface buoys, which can provide precise time series but fail to capture the spatial structure of wave fields. Researchers have explored multi-point measurement arrays to address this limitation. For instance, Swan and Sheikh (2015) deployed 1400 probes to map the spatial–temporal wave fields. However, this approach faces practical challenges due to operational complexity and required repeated trials, and the instruments may interfere with wave dynamics, potentially introducing measurement noise. Moreover, Benetazzo et al. (2015) found that point-measurement-based models inadequately predict certain wave characteristics, particularly extreme waves (Benetazzo et al., 2021). X-band radar is a key remote sensing device for large-scale ocean wave measurement (Kusters et al., 2016; Naaijen et al., 2018), utilizes empirical Modulation Transfer Functions (MTFs) to extract wave field data from radio wave backscatter (Benetazzo et al., 2018). Klein et al. (2020) and Fujimoto and Waseda (2020) proposed non-linear models to fit the wave field, but in practice the process requires long calibration periods and additional wave measurement equipment.

Therefore, there is a need to develop reliable and efficient methods to measure spatial–temporal wave fields. In recent years, binocular stereo vision has shown great potential in this field, finding widespread application in both outdoor water surfaces (Benetazzo, 2006; Benetazzo et al., 2012) and laboratory wave tanks (Bechle and Wu, 2011; Li

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et al., 2022) for wave field reconstruction. The success of this method primarily depends on the quality of 3D point clouds, which directly relies on accurate pixel matching between stereo image pairs. Lavieri et al. (2020) demonstrated that in outdoor environments, accurate matching can be achieved with sufficient image texture when cameras are positioned at proper distances from the water surface. In general, the sky provides a uniform background light source, allowing the sea surface to behave as a Lambertian surface.

The Wave Acquisition Stereo System (WASS) represents a significant advancement in stereo vision wave measurement, being an open-source software developed by Bergamasco et al. (2017) that combines modern computer vision techniques to provide an effective tool for dense reconstruction of ocean wave. The system encompasses the entire pipeline of generating 3D point clouds using Semi-Global Block Matching (SGBM) as introduced by Hirschmuller (2007), extracting sea level and elevation data and rendering ocean wave height onto images. WASS has been widely utilized for wave characterization in various environments such as the sea surface (Guimarães et al., 2019; Vieira et al., 2020) and laboratory wave tanks (Benetazzo et al., 2019; Pereira et al., 2021).

However, wave reconstruction remains computationally intensive, as dense reconstruction requires processing vast amounts of pixel data, reducing efficiency in both stereo image matching and subsequent processing including smoothing interpolation and outlier filtering. To break through this bottleneck, Bergamasco et al. (2021) proposed WASSfast, which aims at real-time wave field reconstruction using sparse 3D points. The method transforms the spectrum estimation into a constraint minimization problem, seeking an optimal balance between the predicted spectrum and the elevation from the sparse 3D points. To avoid overfitting from strict constraints, the soft constraint variant was also explored in WASSfast. WASSfast employs an iterative gradient descent instead of the closed-form solutions due to the ill-conditioned matrices caused by data sparsity. Additionally, Pistellato et al. (2021) developed a CNN mode of WASSfast that integrates physics prediction with network architecture, trained on fixed 256×256 grids for direct sparse-to-dense elevation mapping.

WASSfast not only establishes a new paradigm for wave field estimation, but also enables reinterpretations within different theoretical frameworks. From Bayesian perspectives, fusing prediction and measurement data represents maximum a posteriori (MAP) estimation, where the prediction model provides the prior and wave measurements supply likelihood data. This MAP problem is solvable recursively via Kalman filter (KF), which can estimate state uncertainty using iterated Fisher information matrices. Indeed, researchers have explored wave field estimation using KF. For instance, Pascoal et al. (2017) and Kim (2020) proposed Kalman filter-based approaches for sea state estimation, but using multiple ships concurrently as buoys. They constructed the state vectors from wave components (paired real and imaginary spectrum parts) evolving independently, typically using tens of entry frequencies.

In stereo vision-based wave reconstruction, although point clouds provide better spatial representation of wave fields, the high dimensionality ($O(10^4)$ in this study) of the state vector flattened from the 2D wave elevation grid makes direct application of the Kalman filter impractical. To address this, we propose a spatial–temporal wave field estimation method based on the ensemble transform Kalman filter (ETKF). As an extension of the ensemble Kalman filter (EnKF), ETKF has been widely used in large-scale data assimilation, particularly for weather and ocean forecasting (Blockley et al., 2014; Zhou et al., 2019). By maintaining an ensemble of states and using transform techniques to avoid large-scale matrix inversions, the proposed ETKF-based method significantly enhances processing efficiency. The proposed method adopts WASSfast's prediction model, and interpolates sparse 3D points to a uniform grid of wave elevations as observations. The main contributions of this paper are: (1) A theoretical comparison between the frameworks of WASSfast's constrained optimization and MAP-based recursive Kalman filtering, under the paradigm of fusing spectrum prediction with sparse stereo 3D reconstruction; (2) An ETKF implementation for spatial–temporal wave field estimation with optical sparse measurement, integrating recursive processing and fuzzy logic to achieve efficient computation and outlier elimination.

The rest of this paper is organized as follows: Section 2 reviews the theoretical framework of the new paradigm in WASSfast, and examines the fundamental connection between the constrained optimization problem and the Kalman filter framework. Section 3 details the proposed fuzzy logic-enhanced ETKF method for estimating spatial–temporal wave fields. Section 4 evaluates the method using a public stereo image dataset of ocean waves (Guimarães et al., 2020) as well as the field-collected data in a vessel on position keeping. Finally, conclusions are given in Section 5.

2. Comparative analysis

This study begins with a brief overview of WASSfast, a paradigm combines the sparse 3D point cloud with wave spectrum prediction and updating. We aim to investigate the theoretical basis of the proposed method by analyzing the relationship between the cost functions of the Kalman filter and the constrained optimization in WASSfast.

2.1. The WASSfast paradigm

The spectrum prediction utilizes the linear dispersion relation of gravity waves. Given a 2D wave number spectrum $S(K_x, K_y)_{t_0}$ at time t_0, the prediction steps are as follows:

1. $S(K_x, K_y)_{t_{n-1}}$ is obtained from $Z(X, Y, t_{n-1})$ using 2D Fast Fourier Transform (FFT), where $Z(X, Y, t_{n-1})$ represents the 2D grid of reconstructed wave heights at time t_{n-1}.
2. For each wave number (K_x, K_y), compute the angular frequency ω_{k_x,k_y}:
$$\omega^2_{k_x,k_y} = \text{sign}(s_x k_x + s_y k_y) g \sqrt{k_x^2 + k_y^2} \tanh(d\sqrt{k_x^2 + k_y^2}) \tag{1}$$
where g is the acceleration of gravity, d is the depth of water, and s_x and s_y are the signs of the wave propagation direction.
3. Rotate the phase of $S(K_x, K_y)_{t_{n-1}}$ to obtain the predicted spectrum $S'(K_x, K_y)_{t_n}$ at moment t_n:
$$\begin{aligned} S'\left(K_x, K_y\right)_{t_n} &= S\left(K_x, K_y\right)_{t_{n-1}} e^{i(t_n - t_{n-1})\omega_{k_x,k_y}} \\ &\triangleq \mathcal{S}_{t_n|t_{n-1}} \end{aligned} \tag{2}$$

This prediction equation provides a simplified model of the sea state, neglecting non-linear effects and the Doppler effect of currents. However, it serves as a reasonable initial estimate for subsequent spectrum updating. The updating process aims to fuse the predicted spectrum S' with the current 3D point cloud Γ_{t_n}, reconstructed using a feature matching method based on optical flow (Shi and Tomasi, 1994; Bouguet, 2001). The spectrum updating is formulated as a minimization problem with constraints:

$$\begin{cases} \arg\min_S w\|S - S'\| \\ s.t. \qquad \mathbf{A}S = b \end{cases} \tag{3}$$

where, S is the spectrum to be optimized, S' is the predicted spectrum, w is a flattened (row-wise) high-pass Gaussian window of size $N \times N$ with a certain cutoff, used to apply greater weights in the high-frequency part of the spectrum, $\mathbf{A}$ is the inverse discrete Fourier transform (iDFT) matrix of $B \times N^2$, B is the number of sparse points, and b is a vector of sparse elevations.

Applying the Karush–Kuhn–Tucker (KKT) condition, the multipliers vector is $\lambda = [\lambda_1 \ldots \lambda_M]$, and since $\|\cdot\|$ is close in the L2 sense, the cost function can be rewritten as:

$$L(S, \lambda) = \left(S - S'\right)^H \mathbf{W}\left(S - S'\right) + \lambda^T(\mathbf{A}S - b) \tag{4}$$

where $\mathbf{W} = \omega\mathbf{I}$. The goal of Eq. (4) is to find a new spectrum S that is as close as possible to the predicted spectrum S' (especially in the high-frequency part) with the dimension $N^2 \times 1$, and to make sure that the grids obtained by iDFT accurately interpolate all the sparse 3D points.

By taking partial derivatives of $L(S, \lambda)$ with respect to S and λ and equating them to zero, one obtains:

$$\begin{cases} 2\mathbf{W}(S - S') - \mathbf{A}^H \lambda = 0 \\ \mathbf{A}S = b \end{cases} \tag{5}$$

The matrix form of the Eq. (5) can be expressed as:

$$\begin{pmatrix} \mathbf{W} & \frac{\mathbf{A}^H}{2} \\ \mathbf{A} & 0 \end{pmatrix} \begin{pmatrix} S \\ \lambda \end{pmatrix} = \begin{pmatrix} \mathbf{W}S' \\ b \end{pmatrix} \tag{6}$$

Finally, using conjugate gradient iteration, x is updated to find the residual minima between $[\mathbf{W}S', b]^T$ and $\mathbf{F}x$, where $\mathbf{F}x$ is $[\mathbf{W}S + \mathbf{A}^H\lambda/2, \mathbf{A}S]^T$, $x = [S, \lambda]^T$.

The NFFT variant (CPU implementation) of WASSfast requires accurate interpolation of sparse points as a strict constraint. Given that the reconstructed elevations from sparse matches are non-uniformly distributed on a 2D grid, NFFT is used to accelerate the computation of $\mathbf{A}^H\lambda$ and $\mathbf{A}S$. In the GPU-accelerated implementation, the strict constraint $\mathbf{A}S = b$ is relaxed, i.e., the estimated spectrum are no longer required to accurately reconstruct the sparse points. This can be expressed by the cost function in Eq. (7):

$$\begin{aligned} L(S) = &\left(S - S'\right)^H \mathbf{W} \left(S - S'\right) \\ &+ \frac{\dot{\alpha}}{B} \left(\mathbf{A}_{\mathcal{F}^{-1}} S - b_B\right)^T \left(\mathbf{A}_{\mathcal{F}^{-1}} S - b_B\right) \end{aligned} \tag{7}$$

where $\dot{\alpha}$ is a scalar weight balancing data constraint strength, $\mathbf{A}_{\mathcal{F}^{-1}}$ is the inverse Fast Fourier Transform, and b_B is the elevation vector from interpolated uniform grid. The second term of the optimization function is no longer strictly required to match the data points exactly.

2.2. The Kalman filter framework

Consider the spatial–temporal wave field as a dynamical system,

$$\begin{aligned} \mu_{t_n} &= \mathbf{A}\mu_{t_{n-1}} + \sigma_{t_n} \\ z_{t_n} &= \mathbf{H}\mu_{t_n} + \upsilon_{t_n} \end{aligned} \tag{8}$$

where $\sigma_t \sim \mathcal{N}(0, \mathbf{Q})$ and $\upsilon_t \sim \mathcal{N}(0, \mathbf{R})$ are time-independent and mutually independent, and $\mu_{t_0} \sim \mathcal{N}(\bar{\mu}_{t_0}, \mathbf{P}_{t_0})$ is independent of σ_{t_0} and υ_{t_0}. Here, the state μ represents the spectrum S, while $\mathbf{A}$ describes a one-step prediction based on the linear dispersion relation of gravity waves. $\mathbf{Q}$ and $\mathbf{R}$ denote process and observation noise, respectively, and $\mathbf{H}$ is the observation or measurement matrix.

Given a set of measurements $z_{t_0}, z_{t_1}, \ldots, z_{t_{N-1}}$ we need to estimate $\mu_{t_0}, \mu_{t_1}, \ldots, \mu_{t_N}$. Therefore, the maximum a posteriori probability (MAP) is estimated to be

$$(\hat{\mu}_t)_{t_0}^{t_N} \in \arg \max_{\mu_{t_0}, \ldots, \mu_{t_N}} p(\mu_{t_0}, \ldots, \mu_{t_{N-1}} | z_{t_0}, \ldots, z_{t_{N-1}}) \tag{9}$$

So based on Bayes rules and the assumption of the first-order Markovianity, we have the relationship in Eq. (10),

$$p\left(\mu_{t_0}, \ldots, \mu_{t_N} \mid z_{t_0}, \ldots, z_{t_{N-1}}\right) \propto p_{\mu_{t_0}}(\mu_{t_0}) \prod_{t_1}^{t_{N-1}} p_{\upsilon_{t_n}}(\upsilon_{t_n}) p_{\sigma_t}(\sigma_{t_n}) \tag{10}$$

where the proportional forms of probability density functions (PDF) are as follows,

$$\begin{aligned} p_{\sigma_{t_n}}(\sigma) &\propto \exp\left(-\frac{1}{2}\sigma^T \mathbf{Q}^{-1} \sigma\right) \\ p_{\upsilon_{t_n}}(\upsilon) &\propto \exp\left(-\frac{1}{2}\upsilon^T \mathbf{R}^{-1} \upsilon\right) \\ p_{\mu_{t_0}}(\mu_{t_0}) &\propto \exp\left(-\frac{1}{2}(\mu_{t_0} - \bar{\mu})^T \mathbf{P}_{t_0}^{-1} (\mu_{t_0} - \bar{\mu})\right) \end{aligned} \tag{11}$$

Taking logarithms of the Eq. (10) yields the log-likelihood in Eq. (12). The cost function for the maximum a posteriori probability (MAP) estimation problem is posed as in Eq. (13).

$$\begin{aligned} &\ln p(\mu_{t_0}, \ldots, \mu_{t_N} | z_{t_0}, \ldots, z_{t_{N-1}}) \\ &= \ln p_{\mu_{t_0}}(\mu_{t_0}) + \sum_{t_1}^{t_{N-1}} \left(\ln p_{\upsilon_t}(z_{t_n} - \mathbf{H}\mu_{t_n}) + \ln p_{\sigma_t}(\mu_{t_n} - \mathbf{A}\mu_{t_{n-1}})\right) \\ &= -\frac{1}{2}(\mu_{t_0} - \bar{\mu}_{t_0})^T \mathbf{P}_{t_0}^{-1} (\mu_{t_0} - \bar{\mu}_{t_0}) \\ &\quad + \sum_{t_0}^{t_{N-1}} \left(-\frac{1}{2}(z_{t_n} - \mathbf{H}\mu_{t_n})^T \mathbf{R}^{-1} (z_{t_n} - \mathbf{H}\mu_{t_n})\right. \\ &\quad \left. - \frac{1}{2}(\mu_{t_n} - \mathbf{A}\mu_{t_{n-1}})^T \mathbf{Q}^{-1} (\mu_{t_n} - \mathbf{A}\mu_{t_{n-1}})\right) \end{aligned} \tag{12}$$

$$\begin{cases} \min\limits_{\mu_{t_0}, \ldots, \mu_{t_N}, \sigma_{t_1}, \ldots, \sigma_{t_N}, \upsilon_{t_1}, \ldots, \upsilon_{t_N}} \frac{1}{2}(\mu_{t_0} - \bar{\mu}_{t_0})^T \mathbf{P}_{t_0}^{-1} (\mu_{t_0} - \bar{\mu}_{t_0}) \\ \qquad + \sum_{t_0}^{t_{N-1}} \left(\frac{1}{2}{\upsilon_{t_n}}^T \mathbf{R}^{-1} \upsilon_{t_n} + \frac{1}{2}{\sigma_{t_n}}^T \mathbf{Q}^{-1} \sigma_{t_n}\right) \\ s.t. \quad \mu_{t_n} = \mathbf{A}\mu_{t_{n-1}} + \sigma_{t_n} \\ \qquad\quad z_{t_n} = \mathbf{H}\mu_{t_n} + \upsilon_{t_n} \end{cases} \tag{13}$$

In Eq. (13), the objective is to estimate a state μ_{t_0} that minimizes its distance from its expected value $\bar{\mu}_{t_0}$ under the weighting of $\mathbf{P}_{t_0}^{-1}$, while maintaining the states μ_{t_n} as close as possible to both predictions and observations. The weighting matrices in the cost function are determined by the inverse covariance matrices, where larger covariance matrices allow larger noise terms. Al-Matouq et al. (2013) showed that this objective function in Eq. (13) can be solved using dynamic programming, resulting in the five Kalman filter equations presented in Eq. (14) and (15).

The prediction steps are shown in Eq. (14),

$$\begin{aligned} \mu_{t_n|t_{n-1}} &= \mathbf{A}\hat{\mu}_{t_{n-1}|t_{n-1}} \\ \mathbf{P}_{t_n|t_{n-1}} &= \mathbf{A}\hat{\mathbf{P}}_{t_{n-1}|t_{n-1}}\mathbf{A}^T + \mathbf{Q} \end{aligned} \tag{14}$$

and the update steps of KF are shown in Eq. (15),

$$\begin{aligned} \mathbf{K} &= \mathbf{P}_{t_n|t_{n-1}}\mathbf{H}^T \left(\mathbf{H}\mathbf{P}_{t_n|t_{n-1}}\mathbf{H}^T + \mathbf{R}\right)^{-1} \\ \hat{\mu}_{t_n|t_n} &= \mu_{t_n|t_{n-1}} + \mathbf{K}\left(z_{t_n} - \mathbf{H}\mu_{t_n|t_{n-1}}\right) \\ \hat{\mathbf{P}}_{t_n|t_n} &= (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_{t_n|t_{n-1}} \end{aligned} \tag{15}$$

The above analysis in this section suggests that the cost function in MAP estimation considers global information and can be solved recursively using the KF, where the recursive process propagates the uncertainty from the previous time step to the current one, contributing to optimal state estimation. This propagation compresses historical information into a dynamic covariance matrix, reflecting the estimation uncertainty of the corresponding dimension. In contrast, the optimization problem in Eq. (4) and (7) employ a fixed weight matrix $\mathbf{W}$ as a frequency domain filter to penalize high-frequency deviations, which reflects a prior smoothness assumption, and uses a scalar $\dot{\alpha}$ to balance prior information and observed data.

3. Methodology

This section describes a methodology for ocean wave reconstruction using the Ensemble Transform Kalman Filter (ETKF) framework. The proposed method employs sparse 3D reconstruction to provide measurements for an enhanced ETKF implementation, which incorporates fuzzy logic for robust outlier handling. The measurement system and filtering mechanism are presented.

3.1. The measurement in Kalman filter

3.1.1. Sparse reconstruction

The accurate 3D sparse reconstruction of ocean wave surfaces provides measurements for the Kalman filter over the defined grid area.

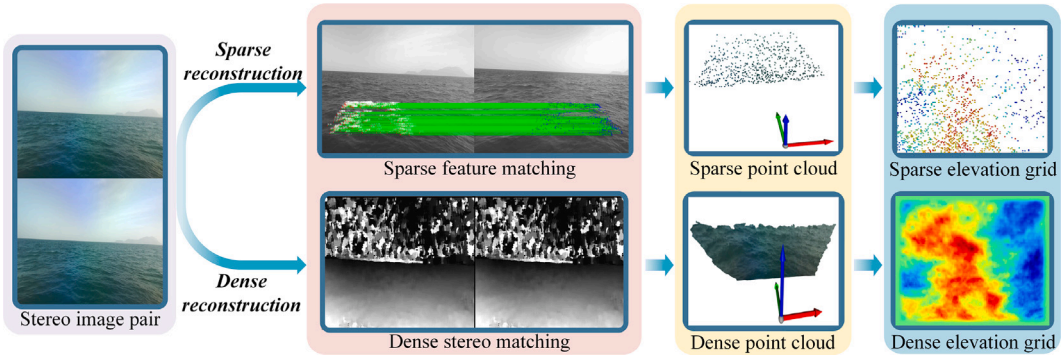


Fig. 1. Comparison of sparse and dense stereo reconstruction pipelines for ocean wave surface, showing the processing steps and resulting elevation grids.

Our goal is to efficiently obtain sparse 3D points through precise feature matching between stereo image pairs. This feature-based approach allows us to focus computational resources on the most distinctive and reliable points of the water surface, rather than attempting to match every pixel. Fig. 1 shows the process of wave field reconstruction using sparse and dense matching points, respectively.

Firstly, feature detection is performed using the Shi-Tomasi corner detector (Shi and Tomasi, 1994) to identify salient corner features from image pairs. These features are typically located at wave crests, troughs, and regions with significant intensity variations on the water surface. In practice we found that this corner-based approach achieves a good balance between computational efficiency and feature stability compared to more complex detectors.

The second step involves feature matching. The Lucas–Kanade optical flow (Bouguet, 2001) is utilized to track the detected features, which has demonstrated robust performance in handling water surface textures. To ensure matching reliability, feature matches with large reprojection errors are eliminated. After filtering, thousands of reliable feature correspondences are typically obtained.

The final two steps are 3D reconstruction and mapping to a 2D grid. Using the camera parameters calibrated through the method by Zhang (1999), we triangulate the matched features to obtain 3D coordinates. During this process, we further apply geometric constraints to filter out physically implausible points. For instance, considering the measurement range and camera mounting height, the reconstructed points should fall within reasonable elevation bounds.

3.1.2. Interpolated measurements

With sparse observation points available, two measurement strategies are considered within the KF framework: direct assimilation of sparse values and assimilation of linearly interpolated values. Both options are examined in the experiments section. For the interpolated measurements, we adopt a linear interpolation method based on Delaunay triangulation, which achieves a good balance between computational efficiency and interpolation accuracy. Specifically, for a target point located within a triangle formed by vertices $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, and $C(x_3, y_3, z_3)$, the interpolated height z is computed as:

$$z = w_1 z_1 + w_2 z_2 + w_3 z_3 \tag{16}$$

where weights w_1, w_2, w_3 are barycentric coordinates satisfying $w_1 + w_2 + w_3 = 1$. The valid points (typically fewer than 7000) are distributed highly non-uniformly on 128 × 128 or 256 × 256 grids, leading to point clusters and data gaps across the surface. Consequently, the interpolation of measurements with sparse and non-uniform distribution inevitably introduces noise and outliers. As illustrated in Fig. 3, comparing the interpolated map derived from sparse points with the map reconstructed from dense points shows the outliers introduced in the interpolation.

Then we analyze state increments to investigate how measurements influence the propagation between different dimensions of the state. Consider an n-dimensional state space: $\boldsymbol{\mu} \in \mathbb{R}^n$, and an m-dimensional innovation vector (measurement space): $\mathbf{d} \in \mathbb{R}^m$ ($m \leq n$). For any state dimension i, the state increment $\Delta\mu_i$ can be written as:

$$\Delta\mu_i = \sum_{j=1}^{m} K_{ij} d_j \tag{17}$$

where d_j represents the jth dimension of $\mathbf{d}$, and K_{ij} is an element of the Kalman gain matrix, which can be expressed as:

$$K_{ij} = \sum_{k=1}^{m} p_{ik} \left(\mathbf{S}^{-1}\right)_{jk} \tag{18}$$

The matrix $\mathbf{S}$, which serves as the normalization denominator in measurement space, is given by:

$$\mathbf{S} = \mathbf{HPH}^T + \mathbf{R} \tag{19}$$

where $\mathbf{H} \in \mathbb{R}^{m \times n}$ is the observation matrix that provides a linear transformation from state space to measurement space. Each row i of $\mathbf{H}$ consists of zeros except for a value 1 at column j, where j corresponds to the directly measured state variable.

Due to the sparse structure of $\mathbf{H}$, only the covariances between state dimensions corresponding to the measurement space contribute to $\mathbf{S}$. Let $O = \{o_1, o_2, \ldots, o_m\}$ denote the set of state dimensions corresponding to non-zero entries in $\mathbf{H}$, the state increment for any dimension i can then be written as:

$$\Delta\mu_i = f\left(\left\{d_j\right\}_{j=1}^{m}, \left\{p_{o_k o_l}\right\}_{k,l=1}^{m}, \left\{p_{i o_k}\right\}_{k=1}^{m}, \mathbf{R}\right) \tag{20}$$

where $\left\{p_{o_k o_l}\right\}_{k,l=1}^{m}$ denotes the set of covariances between state dimensions corresponding to the measurement space, and $\left\{p_{i o_k}\right\}_{k=1}^{m}$ is the set of covariances between any state dimension i and all directly observed state dimensions corresponding to the measurements.

Eq. (20) indicates that increasing the measurement dimension can expand the set of state covariance $\left\{p_{o_k o_l}\right\}_{k,l=1}^{m}$ and $\left\{p_{i o_k}\right\}_{k=1}^{m}$ participating in updates. By increasing measurement dimensions, this expansion not only contributes to directly observed state dimensions, but also enriches the set of correlations utilized in updating unobserved states. These correlations are fundamentally established by the prediction model, which encodes prior physical constraints into the covariance values during the prediction step. In practice, state localization is typically employed to retain significant local correlations. Therefore, the interpolated measurements enable more comprehensive exploitation of state covariance and better capture of spatial correlations during updates. The artifacts introduced in the interpolation procedure are inherently accommodated through the uncertainty-aware weighting of the Kalman filter framework.

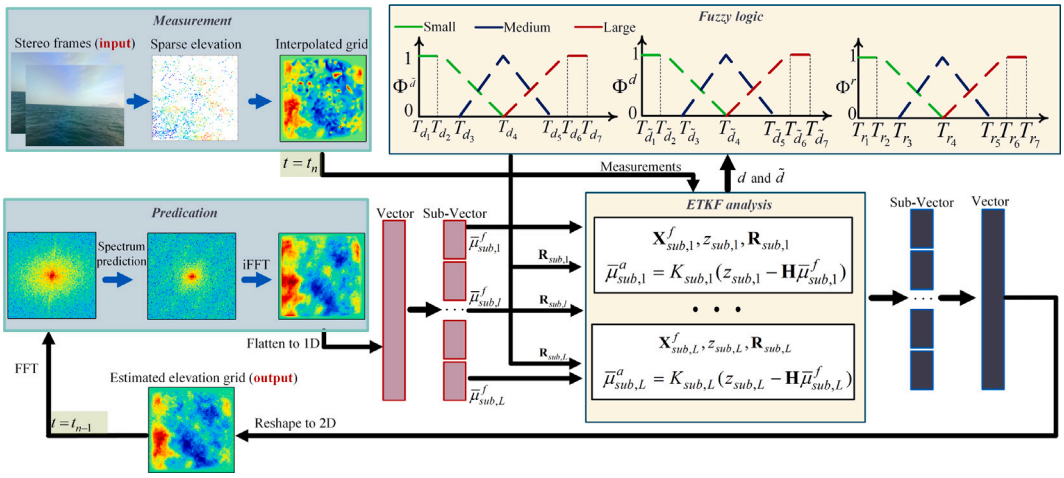


Fig. 2. The Fuzzy logic-enhanced ETKF-based reconstruction pipeline. The prediction step transforms the previous time step's estimated elevation grid to a spectrum by FFT, applies linear gravity wave dispersion relations for spectrum prediction, and converts back to 2D elevation by iFFT. Then the 2D elevation is flattened into a 1D state vector and localized into L sub-state vectors for ETKF analysis. Observations or measurements are obtained on an interpolation grid of sparse reconstruction. The innovation (difference between observation and state prediction) and its change from the previous time step are input into a fuzzy logic system, which outputs adaptive measurement noise sub-matrices for each sub-state. After the analysis of ETKF, sub-state vectors are combined into a complete state vector and reshaped to 2D, yielding the current time step's estimation.

3.2. Ensemble transform Kalman filter

When performing actual calculations, the equations for one-step state transition and the predicted measurement vector are as follows,

$$\begin{aligned} \mu^f_{t_n} &= \mathcal{A}\left(\mu^a_{t_{n-1}}\right) \\ z^f_{t_n} &= \mathbf{H}\mu^f_{t_n} \end{aligned} \tag{21}$$

where the state μ_{t_n} is defined as the flatten vector of the elevation grid at time t_n with dimension $1 \times N^2$. σ_{t_n} and v_{t_n} represent the prediction model error vector and the observation error vector, respectively. The prediction on the state is denoted as $\mathcal{A}\left(\mu_{t_{n-1}}\right) = \mathcal{F}^{-1}\left(S_{t_n t_{n-1}}\left(\mathcal{F}\left(\mu_{t_{n-1}}\right)\right)\right)$, where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the fast Fourier transform (FFT) and inverse transform (iFFT), respectively. The observation matrix $\mathbf{H}$ is the identity matrix in the case of assimilating interpolated measurement.

There are two main reasons for adopting elevation rather than spectrum as the state. Firstly, minor numerical perturbations in the frequency domain can lead to significant errors when reconstructing elevation within the filtering. Secondly, using elevation simplifies the observation matrix $\mathbf{H}$ to an identity matrix, reducing computational cost compared to the tensor product required for a spectrum state. Nevertheless, the spectrum remains an intermediate state in the prediction step, as illustrated in Fig. 2. However, since the state space in this study exhibits high dimensionality, the standard KF is computationally expensive and requires excessive storage for practical implementation, primarily due to the propagation of large forecast error covariance matrices.

Therefore, ETKF is utilized to estimate the wave space–time field in this study, which is a Monte Carlo implementation of the Bayesian estimate problem and was first proposed for use in the atmospheric sciences by Bishop et al. (2001). Suppose the ensemble size is M, and the state $\mu^a_{t_{n-1}}$ are perturbed to generate the ensemble $\mu^a_{t_{n-1},j}$, $j = 1, \ldots, M$. The Ensemble of states is $\mathbf{X}^a_{t_{n-1}}\left(\mu^a_{t_{n-1},1}, \mu^a_{t_{n-1},2}, \ldots, \mu^a_{t_{n-1},M}\right)$, and the forecast ensemble is $\mathbf{X}^f_{t_n}\left(\mu^f_{t_n,1}, \mu^f_{t_n,2}, \ldots, \mu^f_{t_n,M}\right)$, in which the forecast state is defined as given in Eq. (22),

$$\mu^f_{t_n,j} = \mathcal{A}\left(\mu^a_{t_{n-1},j}\right) \tag{22}$$

The mean forecast state $\bar{\mu}^f_{t_n}$ is defined as the ensemble mean of $\mu^f_{t_n,j}$, i.e.,

$$\bar{\mu}^f_{t_n} = \frac{1}{M}\sum_{j=1}^{M}\mu^f_{t_n,j} \tag{23}$$

The perturbation ensemble of the predicted state ensemble is obtained by subtracting the ensemble mean from the state ensemble,

$$\mathbf{Y}^f_{t_n} = \frac{1}{\sqrt{M-1}}\left(\mu^f_{t_n,1} - \bar{\mu}^f_{t_n}, \mu^f_{t_n,2} - \bar{\mu}^f_{t_n}, \ldots, \mu^f_{t_n,M} - \bar{\mu}^f_{t_n}\right) \tag{24}$$

In ETKF, the forecast of state error covariances is no longer performed explicitly, but the covariances of the ensemble of predicted states are computed directly as follows,

$$\mathbf{P}^f_{t_n} = \frac{1}{M-1}\sum_{j=1}^{M}\left(\mu^f_{t_n,j} - \bar{\mu}^f_{t_n}\right)\left(\mu^f_{t_n,j} - \bar{\mu}^f_{t_n}\right)^T \tag{25}$$

Comparing Eq. (24) with Eq. (25), the relationship between the covariance matrix and the perturbation matrix is obtained as follows,

$$\mathbf{P}^f_{t_n} = \mathbf{Y}^f_{t_n}\left(\mathbf{Y}^f_{t_n}\right)^T \tag{26}$$

According to Eq. (26), the Kalman gain computation is transformed as follows:

$$\begin{aligned} \mathbf{K}_e &= \mathbf{P}^f_e\mathbf{H}^T\left(\mathbf{H}\mathbf{P}^f_e\mathbf{H}^T + \mathbf{R}\right)^{-1} \\ &= \mathbf{Y}^f\left(\mathbf{Y}^f\right)^T\mathbf{H}^T\left(\mathbf{H}\mathbf{Y}^f\left(\mathbf{Y}^f\right)^T\mathbf{H}^T + \mathbf{R}\right)^{-1} \\ &= \mathbf{Y}^f\left(\mathbf{Z}^f\right)^T\left(\mathbf{Z}^f\left(\mathbf{Z}^f\right)^T + \mathbf{R}\right)^{-1} \end{aligned} \tag{27}$$

where $\mathbf{Z}^f_{t_n} = \mathbf{H}\mathbf{Y}^f_{t_n}$ represents the perturbation ensemble of predicted measurement vector. Let $\hat{\mathbf{Z}}^f = \mathbf{R}^{-\frac{1}{2}}\mathbf{Z}^f_{t_n}$, and perform Singular Value Decomposition (SVD) for $(\hat{\mathbf{Z}}^f)^T$, one obtains:

$$\left(\hat{\mathbf{Z}}^f\right)^T = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T \tag{28}$$

where $\mathbf{U}$ is a $m \times m$ orthogonal matrix with m being the dimension of measurement space, $\boldsymbol{\Sigma}$ is a $m \times M$ diagonal matrix where M represents the ensemble size, and $\mathbf{V}$ is a $M \times M$ orthogonal matrix.

Therefore, the actual Kalman gain for the ensemble transformation analysis is:

$$\begin{aligned} \mathbf{K}_e &= \mathbf{Y}^f\left(\hat{\mathbf{Z}}^f\right)^T\left(\hat{\mathbf{Z}}^f\left(\hat{\mathbf{Z}}^f\right)^T + \mathbf{I}\right)^{-1}\mathbf{R}^{-\frac{1}{2}} \\ &= \mathbf{Y}^f\mathbf{U}\boldsymbol{\Sigma}\left(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma} + \mathbf{I}\right)^{-1}\mathbf{V}^T\mathbf{R}^{-\frac{1}{2}} \end{aligned} \tag{29}$$

it can be seen that the transformation effectively reduces the computational complexity since only a diagonal matrix inversion is required. Finally, the mean analysis state are shown in Eq. (30),

$$\bar{\mu}^a_{t_n} = \mathbf{K}_e\left(z_{t_n} - \bar{z}^f_{t_n}\right) \tag{30}$$

Table 1
Fuzzy rules based on innovation and innovations difference.

Rule	IF(d_{t_n})	IF($\tilde{d}_{t_n}$)	THEN(r_{t_n})
1	Large	Large	Large
2	Large	Medium	Large
3	Large	Small	Medium
4	Medium	Large	Large
5	Medium	Medium	Medium
6	Medium	Small	Small
7	Small	Large	Medium
8	Small	Medium	Small
9	Small	Small	Small

where $\bar{\mu}^a_{t_n}$ is the estimated state at time t_n, z_{t_n} is the measurements of wave elevation at time t_n and $\bar{z}^f_{t_n}$ is the mean of predicted measurement ensemble $\mathbf{H}\mathbf{X}^f_{t_n}$.

The transformation matrix and the state perturbation matrix are updated as follows,

$$\begin{aligned}\mathbf{T} &= \mathbf{U}\left(\mathbf{I}+\boldsymbol{\Sigma}\boldsymbol{\Sigma}^T\right)^{-1/2}\mathbf{U}^T \\ \mathbf{Y}^a_{t_n} &= \mathbf{Y}^f_{t_n}\mathbf{T}\end{aligned} \tag{31}$$

The perturbed ensemble of states is computed as follows,

$$\mathbf{X}^a_{t_n} = \overline{\mathbf{X}}_{t_n} + \sqrt{M-1}\mathbf{Y}^a_{t_n} \tag{32}$$

where $\overline{\mathbf{X}}_{t_n}$ is the ensemble consisting of M repetitions of the mean state vector $\bar{\mu}^a_{t_n}$, and $\mathbf{X}^a_{t_n}$ will be predicted in the next iteration at t_{n+1}.

In the above framework of ETKF, it is observed that matrices involving inverse operations are exclusively diagonal, enabling efficient computation through perturbation propagation instead of state covariance matrices.

Of course, it must be said that as a Monte Carlo method, the success of the ETKF greatly depends on ensemble size and statistical representativeness. Lorenc (2003) demonstrated that when ensembles are generated by random sampling via the Probability Density Function (PDF), the discrepancy of forecast error covariance scales inversely with ensemble size M. Furthermore, Wang and Bishop (2003) pointed out that when the number of ensemble perturbations is significantly smaller than the projected directions of forecast error variance, Eq. (31) underestimates the total analytical error variance due to insufficient contributions from crucial parts of the error space. In practical applications, computational constraints often prevent the ensemble size from matching the state dimension (e.g., in this study the state dimension is 16384 × 1 or 65536 × 1, while only tens to hundreds of ensemble members are used). To address this issue, we employ a localization strategy for ETKF, focusing on sub-vectors of states and observations, which can also enable GPU-accelerated parallel processing. Fig. 2 illustrates the localization process. Specifically, the state vector is localized into L sub-states that can be updated in parallel and isolation, where each sub-state vector has a dimension of $(N/L, 1)$. The ensemble size M is typically set to be slightly larger than N/L to ensure adequate sampling. While localized sub-states are used in the analysis step, the forecast step requires a complete state vector corresponding to the elevation grid for FFT operations. The localization shifts the computational bottleneck from matrix inversion to SVD computation. Therefore, the choice of L is a trade-off between parallel SVD efficiency and spatial correlation preservation.

3.3. Fuzzy logic integration

The interpolation of non-uniformly distributed sparse 3D points can introduce significant outliers in the measurement. As shown in Fig. 4 (right), the noise distribution of measurement exhibits heavier tails than a Gaussian distribution. To address this issue, we employ fuzzy logic to adaptively adjust $\mathbf{R}$ by designing appropriate rules based on the current innovation and its difference from the previous one.

The current innovation and the difference of innovations are defined, respectively, as follows,

$$\begin{aligned}d_{t_n} &= z_{t_n} - \bar{z}^f_{t_n} \\ \tilde{d}_{t_n} &= d_{t_n} - d_{t_{n-1}}\end{aligned} \tag{33}$$

Fuzzy logic can distinguish genuine system changes from measurement errors by evaluating innovation and the difference values. High innovation with low difference values suggests a gradual system change, as the observation differs from predictions but remains close to previous data. In this case, the reliability of observation is considered higher. On the other hand, high values in both innovation and difference suggest a significant deviation from both predictions and recent data, potentially indicating an outlier. In such cases, the observation should be assigned a lower confidence. The specific fuzzy rules are shown in Table 1.

For the ith rule, when given inputs (d_{t_n} and $\tilde{d}_{t_n}$), the activation value for the output r_{t_n} is computed as Eq. (34),

$$C_{\text{rule }i}(r_l) = \min\left(\boldsymbol{\Phi}^d\left(d_{t_n}\right), \boldsymbol{\Phi}^{\tilde{d}}\left(\tilde{d}_{t_n}\right), \boldsymbol{\Phi}^r(r_l)\right) \tag{34}$$

where, $\boldsymbol{\Phi}^d(\cdot)$ and $\boldsymbol{\Phi}^{\tilde{d}}(\cdot)$ represent membership functions for the input variables, $\boldsymbol{\Phi}^r(\cdot)$ for the output variables, respectively. Each of the variables is mapped into their membership functions: “small”, “medium”, and “large”. The fuzzy logic is shown in Fig. 2, T_d, $T_{\tilde{d}}$, and T_r are defined as the thresholds in the membership functions of the variables.

The centroid computation yields,

$$\hat{r} = \frac{\sum_{l=1}^{L}\omega(r_l)\cdot r_l}{\sum_{l=1}^{L}\omega(r_l)} \tag{35}$$

where, l=1, 2,..L, and L denotes the number of discrete points for the output variable.

The output variable assigns an adjustment factor to each direction of the $\mathbf{R}$ diagonal,

$$r_{t_n,ii} = \hat{r}_{t_n,ii} \tag{36}$$

where r_{ii} is the ith diagonal element of $\mathbf{R}$.

4. Experiments

We compared the proposed ETKF-based method with the results obtained by WASS and WASSfast for different stereo sequences. The data for the experiment include: the public Acqua Alta stereo data set (AA 03), collected on the Acqua Alta oceanic research platform with a 2.545 m camera baseline and 12.5 m height above sea level, covering a measurement range of −25 m to 25 m in the x-direction and 30 m to 80 m in the y-direction. The detailed setup parameters refer to Guimarães et al. (2020). Additionally, we used a field-collected dataset, gathered in March 2024 at a certain sea area with a 0.12 m camera baseline and approximately 2 m height above sea level, covering −3 m to 3 m in the x-direction and 5 m to 12 m in the y-direction.

4.1. On the public dataset

We first processed all sequences using WASS with the recommended parameters from the publish dataset to perform dense reconstruction as ground truth. The accuracy of WASS dense reconstruction has been extensively validated in previous studies, showing centimeter-level errors when compared with ultrasonic gauge measurements over long-term monitoring periods (Benetazzo, 2006; Benetazzo et al., 2012, 2015). The validation also demonstrated high spectral coherence in frequency domain analyses. Then, we estimated the average sea surface by computing the mean of corresponding surfaces for each frame and defined a surface elevation grid, in which the interpolation was performed. The PU mode of WASSfast based on the discussed constrained optimization was performed on a 128 × 128 grid. For sparse field interpolation, WASSfast also provides a CNN mode that performs direct fitting on a

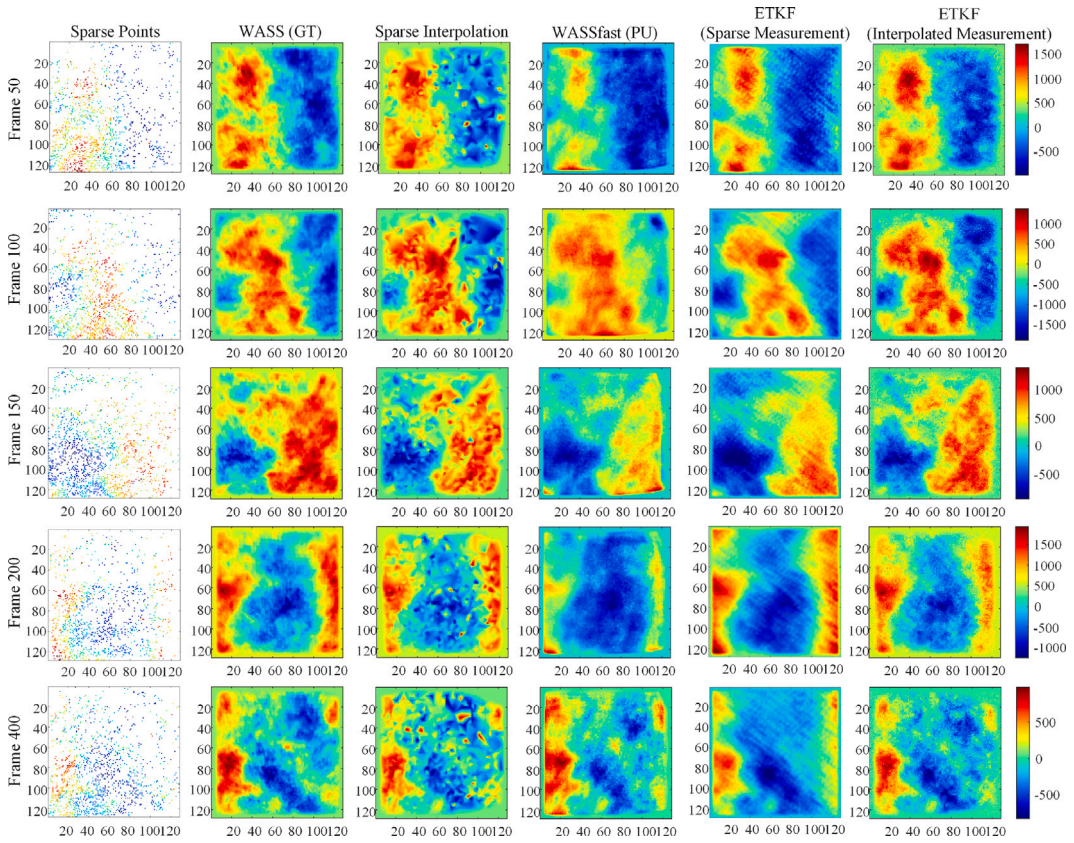


Fig. 3. Wave elevation maps reconstructed using different methods, visualized with color mapping. Each column represents a different reconstruction method: original sparse points, WASS (GT), sparse interpolation, WASSfast (PU), and the proposed ETKF-based method with both sparse measurements and interpolated measurements. The columns display selected frames from the reconstructed 2D elevation grids for each method.

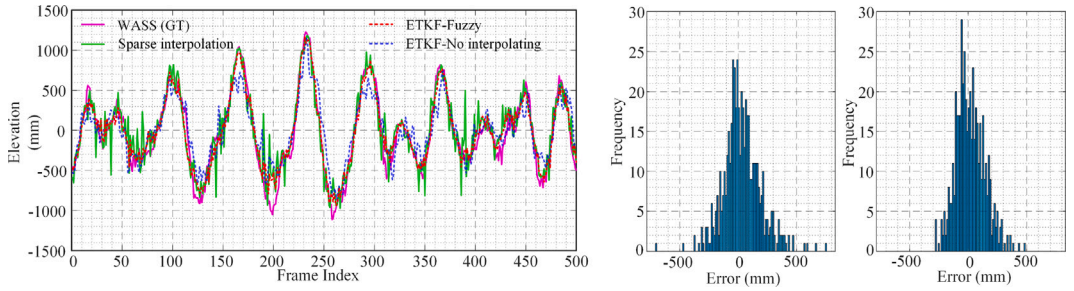


Fig. 4. Evaluation on public dataset. (Left) Comparison of the center grid elevations from ETKF with both sparse and interpolated measurements, the dense reconstruction and sparse interpolation. (Right) Distributions of the absolute error throughout the whole sequences of the center grid in the sparse interpolation and the proposed method, respectively. A dynamic visualization of the wave updating process is provided in the supplementary video.

fixed 256 × 256 grid through a deep neural network. The proposed ETKF-based method was compared with both modes at corresponding grid sizes to evaluate reconstruction accuracy and computational efficiency. All experiments were run on a Linux laptop equipped with an Intel i5-10400 CPU clocked at 2.90 GHz, an Nvidia GeForce GTX 2060, and 32 GB of RAM.

The performance of ETKF with direct assimilation of sparse measurements was first examined. The time series comparisons in Fig. 4 and error curves in Fig. 5(a) indicate higher reconstruction errors compared to ETKF with interpolated measurements. Nevertheless, the results confirm the fundamental effectiveness of the filtering framework with direct assimilation of sparse measurements. As can be seen, sparse interpolation can significantly distort elevation representations on uniform grids due to data gaps and noise caused by keypoint mismatches. The proposed method with interpolated measurements outperforms all compared methods, generating elevation grids with lower MSE. Furthermore, the impact of interpolation on data assimilation was evaluated by tracking errors at both original measurement locations and interpolated points. As can be seen in Fig. 5(b), direct assimilation of sparse measurements in ETKF shows limited improvement at original sparse locations. In contrast, when ETKF assimilates interpolated measurements, it achieves significant error reduction not only at interpolated points but also remarkably at the original sparse locations, demonstrating that interpolated measurements enable better information propagation across state dimensions in ETKF.

Fig. 3 illustrates the qualitative performance of each method in elevation reconstruction. The distribution map shows that the interpolation of sparse data generates sharp discontinuities at data gaps, and measurement outliers can also create local extrema and physically unreasonable values in sparsely sampled regions. ETKF with direct assimilation of sparse measurements can reconstruct the coarse-scale elevation distribution but shows limitations in preserving high-frequency

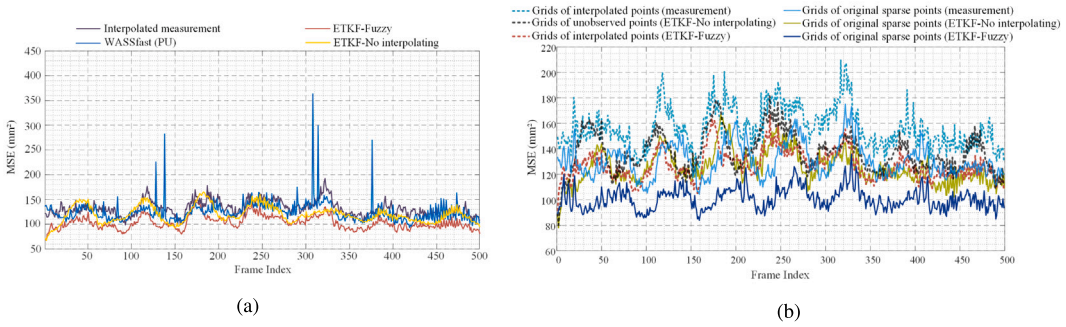


Fig. 5. Evaluation on public dataset. (a) Frame-wise MSE of elevation grids compared to WASS reconstruction. (b) MSE evaluation at original and interpolated grid points comparing sparse reconstruction and ETKF-based method, where ETKF assimilates both sparse and interpolated measurements.

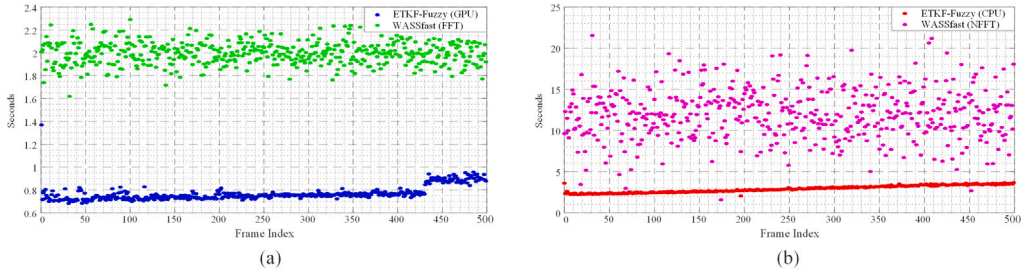


Fig. 6. Evaluation on public dataset. (a) Compute time for the CPU implementation of the proposed method and the NFFT variant of WASSfast. (b) Compute time for the GPU implementation of the proposed method and the FFT variant of WASSfast.

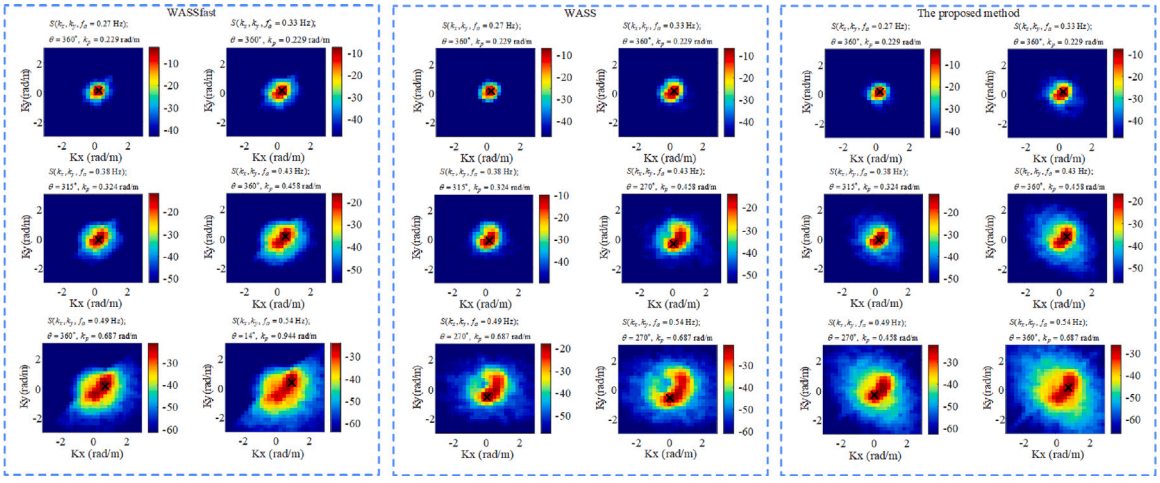
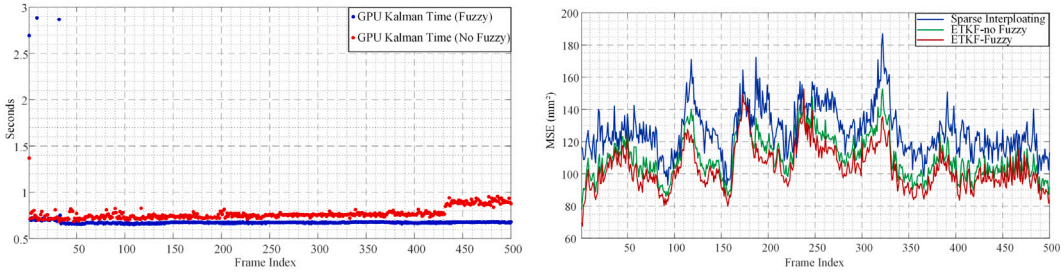


Fig. 7. The figure presents a side-by-side comparison of spectrum analysis results derived from methods: WASSfast (left), WASS (center), and ours (right). Black crosses mark the positions of wavenumber peaks. Each figure displays slices of the three-dimensional spatial–temporal spectrum at selected frequencies, along with the wavenumber and direction represented by the peak point.

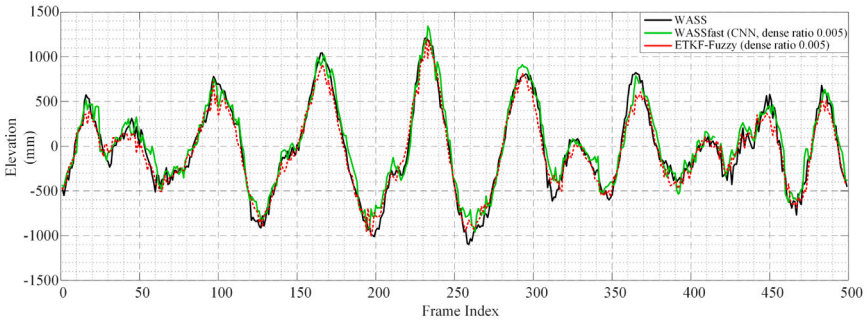
details. WASSfast (PU) tends to smooth the value, suppressing high-frequency information. In contrast, the proposed method preserves more high-frequency details, yielding elevation maps that closely resemble ones obtained from dense reconstruction. Fig. 6 compares the computation time between the proposed method and WASSfast variants using sequence AA 03. The proposed method runs 3–4 times faster on CPU and 2 times faster on GPU compared to WASSfast, demonstrating the computational efficiency of ETKF framework with localization-enabled parallel processing.

Fig. 7 compares the wavenumber spectrum slices of the selected frequencies. The proposed method yields spectrum patterns that are more symmetrical and uniform, closely resembling the results of WASS. In contrast, the spectrum slices of WASSfast show considerable energy loss in the middle and high frequency regions of the second and third quadrants. As demonstrated by Bergamasco et al. (2021), the prediction model assumes that all wavenumber components have the same direction of phase change (increasing or decreasing) in each direction (x or y). This approximation is clearly inconsistent with real ocean waves, in which each wavenumber component can have different phase changes. The prediction model enforces the same sign in the x- and y-directions, leading to non-physical energy redistribution in wavenumber space for components whose predicted phase change direction is opposite to the actual one. The proposed method successfully compensates for this deficiency. Even if the model underestimates or overestimates energy in specific wavenumber regions, corrections can be made by introducing the recursive mechanism of uncertainties (in the form of inverse of the covariance matrix). Moreover, employing fuzzy logic helps prevent outliers.

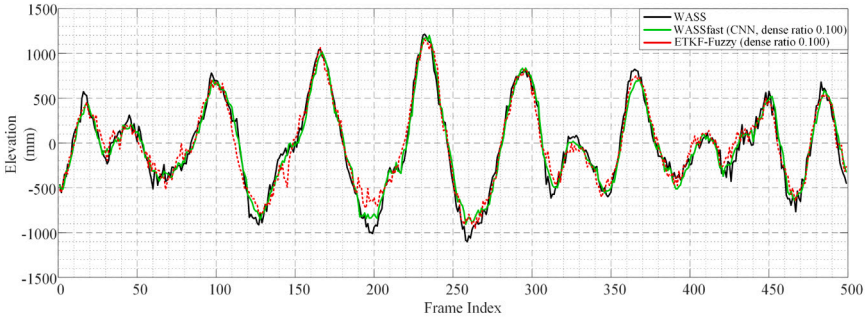


(a) (b)

Fig. 8. Performance of ETKF in Wave Elevation Prediction With and Without Fuzzy Logic Integration. (a) Per-frame GPU computation time for ETKF configurations without and with fuzzy logic. (b) Mean Squared Error (MSE) of 2D elevation grids generated by Sparse Interpolation, ETKF without Fuzzy Logic (ETKF-no Fuzzy), and ETKF with Fuzzy Logic (ETKF-Fuzzy).



(a)



(b)

Fig. 9. Evaluation on public dataset. Comparison of the center grid elevations from the proposed method and CNN mode of WASSfast for dense ratios of (a) 0.005 and (b) 0.100.

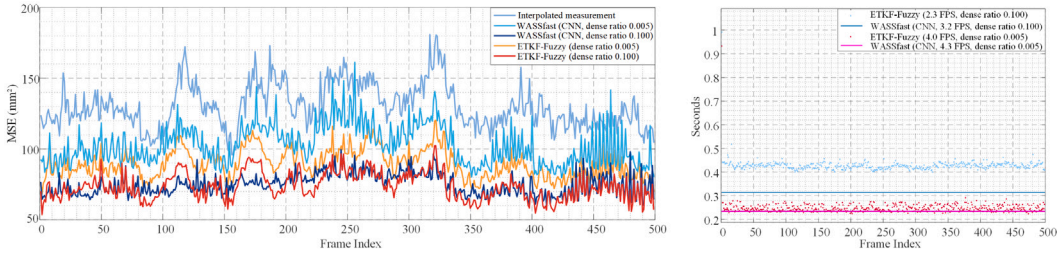
Table 2
Statistics with different ensemble size for the timeseries in the center grid.

Ensemble size	MSE	MSE std	Mean time (ms)
100	23.4	37.7	336
200	17.3	34.8	692
300	14.4	25.5	789
400	16.5	34.8	857
500	16.1	33.9	912

Fig. 8(a) shows that using fuzzy logic in ETKF slightly increases the GPU computation time per frame, but times remain within a close range. Fig. 8(b) compares MSE values, demonstrating that ETKF with fuzzy logic provides the lowest errors and outperforms sparse interpolation and ETKF without fuzzy logic, suggesting the robustness of fuzzy logic in dealing with outliers. From a practical perspective, we believe that the slightly increased computation time is a worthwhile trade-off for the improvement in handling outliers.

Table 2 presents computation times and error statistics for time series generated using ETKF with different ensemble sizes, where the state variable dimension was fixed at 16384 × 1 with 64 sub-states. The results show that while the running time increases with ensemble size, the filter performance does not improve consistently, with optimal performance achieved at an ensemble size of 300. As a matter of fact, this is common in high-dimensional systems within the Monte Carlo-based framework (Bocquet and Sakov, 2013; Poterjoy, 2016), and Sakov and Oke (2008) pointed out that larger size of ensemble in ETKF can sometimes suffer performance degradation due to sampling noise in the observations.

We also evaluated the performance of WASSfast (CNN) and the proposed method on a 256 × 256 grid with two scenarios. A normal case had an average dense ratio of 0.100, while a challenging case used an extremely dense ratio of 0.005 to simulate common challenges in wave imaging, such as poor lighting conditions and white caps. Fig. 9 illustrates the elevation time series reconstructed for two dense ratios. Figs. 10(a) and 10(b) present the detailed MSE comparison and computational time analysis, respectively. WASSfast



(a) (b)

Fig. 10. Evaluation on public dataset. (a) Frame-wise MSE of elevation grids of the proposed method and CNN mode of WASSfast. (b) Compute time for the GPU implementation of the proposed method and CNN mode of WASSfast, where the time for WASSfast (CNN) is from its average frame rate.

Table 3
Computational time analysis of different processing stages within ETKF-Fuzzy.

Processing stage	Dense ratio	
	0.100	0.005
Image Preprocessing	76 ms	76 ms
Point Cloud Generation	185 ms	86 ms
Gridding Interpolation	69 ms	6 ms
State Ensemble Prediction	18 ms	18 ms
Fuzzy Logic Processing	32 ms	32 ms
ETKF State Analysis	45 ms	42 ms

(CNN) demonstrates superior computational efficiency at the higher density ratio and achieves good reconstruction accuracy, validating the effectiveness of its physics-driven architecture and the fast inference capabilities of sparse CNN. Nevertheless, the proposed method shows superior robustness and efficiency under extremely sparse conditions. This can be attributed to the filtering framework that leverages physical correlations between different state dimensions through the covariance structure, which helps maintain the quality of reconstruction even with minimal measurements.

Table 3 presents the detailed mean computational time for different processing stages under varying dense ratios. While image preprocessing and ETKF analysis maintain relatively stable performance, the processing time of point cloud generation and interpolation increases with the density ratio, which provides flexibility in balancing accuracy and computational cost for specific application requirements. The stable ETKF analysis performance is achieved through its recursive structure that enables efficient computation compared to iterative optimization in the Prediction and Update paradigm. Moreover, the ensemble transform reduces matrix inversions to diagonal operations, and state localization facilitates parallel processing.

4.2. On the field-collected data

In order to facilitate future investigations of deck condition based on wave reconstruction, it is necessary to acquire wave images from the vessel (see Fig. 11). A stereo camera ZED2i, was used to capture high-resolution (2208 × 1242) images at 15 frames per second (with an actual frame rate between 10 and 15 frames per second) and with a maximum effective measurement distance of approximately 20 m. The camera is also equipped with an IMU sensor. Throughout the measurement process, the vessel remained stable on the sea surface. Due to frame-to-frame motion in our collected stereo pairs, direct application to the Wass pipeline is not feasible. Following WASS's stereo reconstruction methodology, we employ SGBM stereo matching with similar preprocessing and post-processing steps as in WASS to generate dense point clouds that serve as ground truth. For sparse reconstruction, optical flow is used to generate sparse point clouds.

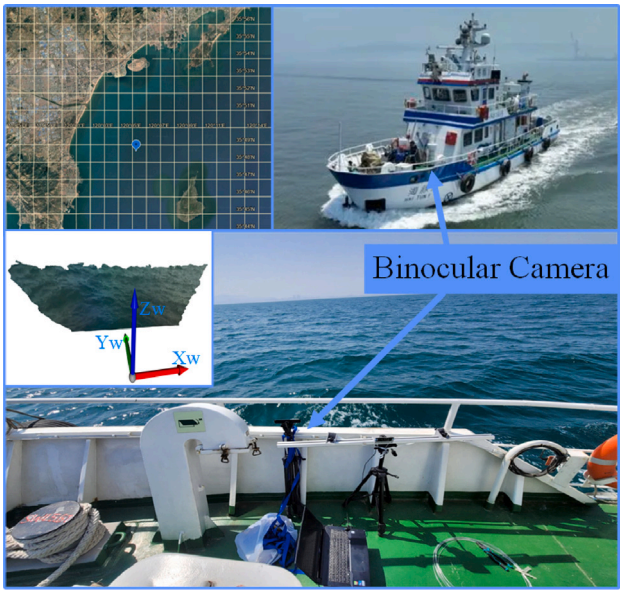


Fig. 11. Experimental setup for on-site data collection. The stereo camera is deployed on the deck of a vessel on position keeping, with the Zw axis of the world coordinate system opposite to the direction of gravity.

We defined the Z-axis of the world coordinate system as opposing the direction of gravity. By capturing the gravity acceleration data corresponding to each frame, it is possible to establish the transformation relationship between the two coordinate systems. The vibration of the vessel can introduce linear acceleration interference, but it is relatively minor compared to the dominant gravitational acceleration signal. Therefore, the acceleration data recorded by the IMU primarily reflects the gravity acceleration acting along the x-, y- and z-axis of the camera coordinate system. The rotation matrix $\mathbf{R}_w$ from the camera coordinate system to the world coordinate system is shown in Eq. (37).

$$\mathbf{R}_w = \mathbf{I} + (\sin\theta)\mathbf{N} + (1 - \cos\theta)\mathbf{N}^2 \tag{37}$$

where $\theta = \arccos\left(\mathbf{z}_c \cdot \mathbf{z}_w\right)$, $\mathbf{N}$ is the skew-symmetric matrix of the rotation axis $\mathbf{n} = \mathbf{x}_c$, $\mathbf{z}_w = -\mathbf{g}$ represents the desired z-axis of the world coordinate system, and $\mathbf{g}$ denotes the unit gravity acceleration vector in the camera coordinate system. $\mathbf{I}$ is a 3x3 identity matrix.

The performance on the field data is shown in Fig. 12 (left), the proposed ETKF-based method successfully filters out the outliers introduced by the interpolation of sparse grid. And Fig. 12 (right) shows the error distributions, one from the proposed method has a shorter tail and fewer outliers beyond ±250 mm, indicating that the update process

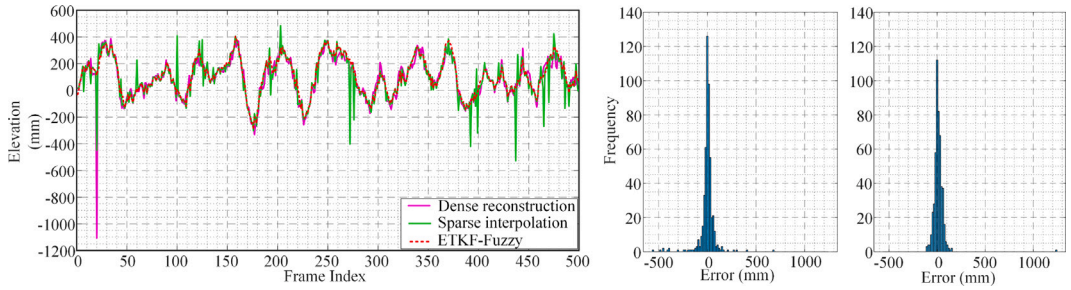


Fig. 12. Evaluation on field-collected data. (Left) Comparison of the center grid elevations from the proposed method, the dense reconstruction and sparse interpolation. (Right) Distributions of the absolute error throughout the whole sequences of the center grid in the sparse interpolation and the proposed method, respectively.

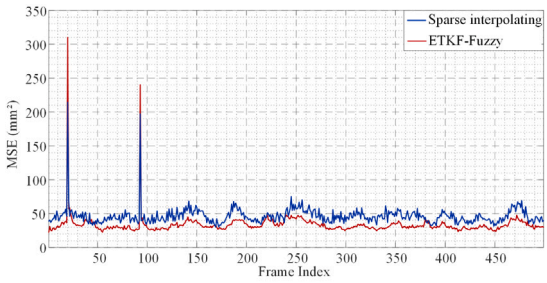


Fig. 13. Evaluation on field-collected data. MSE of 2D elevation grids across the sequence.

effectively reduces the occurrence of large errors, consistent with the performance on the public dataset.

Fig. 13 shows that the proposed method has a significantly lower MSE than the interpolated elevation when both are compared against the dense reconstruction. It is worth noting that significant errors occurred at frames 20 and 90 due to data corruption during collection. We retained the affected frames in the experiment to investigate the robustness of the proposed method in dealing with corrupted measurements, rather than dropping them out. The results demonstrate that following the occurrence of the errors, the method rapidly corrected itself. The qualitative results in Fig. 14 indicates that the proposed method not only maintains consistency with the dense reconstruction results but also significantly enhances the smoothness of elevation between regions.

5. Conclusions

This study proposes an ETKF-based method for fast spatial–temporal wave field reconstruction that combines spectrum predictions with sparse 3D points, using fuzzy logic to filter out measurement outliers. The proposed method inherently achieves global optimization through MAP estimation, distinguishing it from traditional constrained optimization methods with fixed weight matrices. Experiments on both public datasets and field-collected data show superior performance in multiple aspects: single-point elevation timeseries, 2D elevation grids, and 3D spatial–temporal spectrums. The recursive uncertainty propagation chain in ETKF significantly enhances computational efficiency compared to the constrained optimization framework of WASSfast (PU), and the parallel localization of state vector prevents the collapse of covariance matrix in update.

Another important distinction lies in how the two frameworks handle sparse data. While WASSfast directly uses sparse points as support nodes for the fitting surface, the proposed method regards the entire interpolated grid as the measurements, accounting for both the interpolation artifacts and measurement noise of the original sparse points into the observation model within a filter. Through theoretical analysis of innovation propagation and error evaluation, we demonstrate that using interpolated measurements enhances state estimation not only at directly observed locations but across the entire state space while maintaining computational efficiency. The filtering mechanism effectively handles the uncertainty of the interpolation, while leveraging the expanded measurement space to better capture spatial correlations. Furthermore, the proposed method maintains robust reconstruction accuracy and comparable efficiency under extremely sparse conditions, though at the higher dense ratio its computational efficiency is lower than CNN mode of WASSfast. Future work could focus on exploring appropriate entry frequencies as state vectors to reduce state dimensionality, while preserving prior correlations through spectral–spatial transformation in the mode of direct sparse assimilation.

CRediT authorship contribution statement

Feng Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Qidan Zhu:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Chengtao Cai:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Xiaoyu Wang:** Writing – review & editing, Validation, Methodology, Data curation. **Renjie Qiao:** Validation, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.coastaleng.2024.104690.

Data availability

Data will be made available on request. The implementation code is available through Github under https://github.com/fengHEU/Wave-ETKF/tree/main.

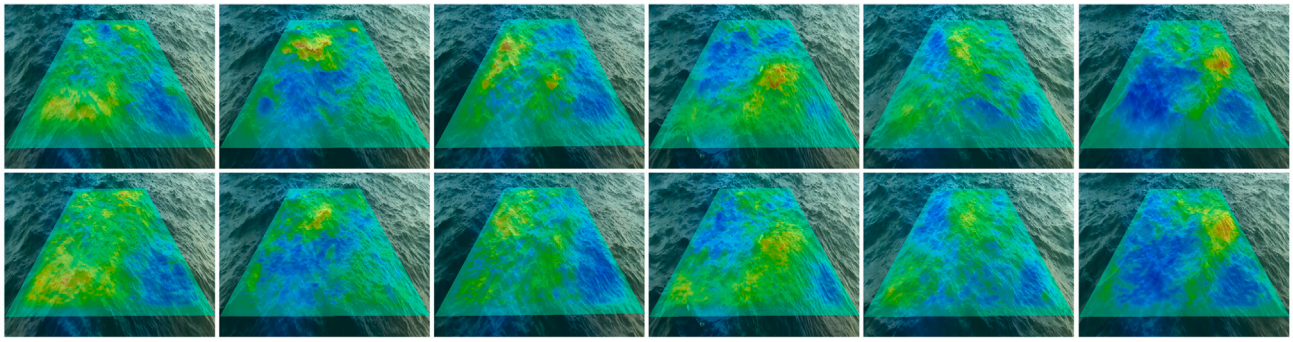


Fig. 14. Elevation rendering on selected images from field-collected data. The top row presents dense wave reconstructions using stereo matching techniques, while the bottom row shows the proposed ETKF-based elevation estimates.

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